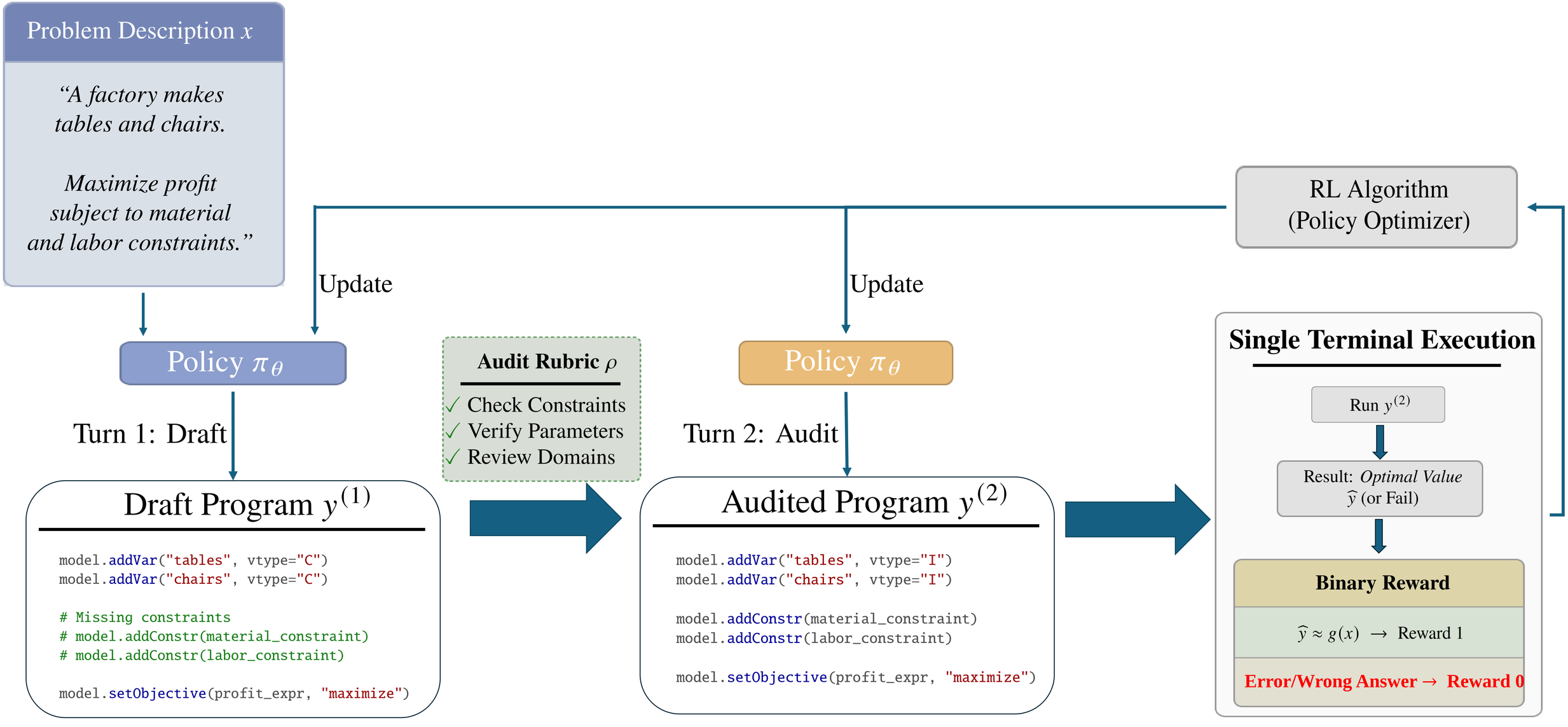
Problem Description x
"A factory makes tables and chairs.
Maximize profit subject to material and labor constraints."
Policy π_θ
Turn 1: Draft
Draft Program $y^{(1)}$
model.addVar("tables", vtype="C")
model.addVar("chairs", vtype="C")
# Missing constraints
# model.addConstr(material_constraint)
# model.addConstr(labor_constraint)
model.setObjective(profit_expr, "maximize")
Update
Audit Rubric ρ
✓ Check Constraints
✓ Verify Parameters
✓ Review Domains
Policy π_θ
Turn 2: Audit
Audited Program $y^{(2)}$
model.addVar("tables", vtype="I")
model.addVar("chairs", vtype="I")
model.addConstr(material_constraint)
model.addConstr(labor_constraint)
model.setObjective(profit_expr, "maximize")
Update
RL Algorithm (Policy Optimizer)
Single Terminal Execution
Run $y^{(2)}$
Result: Optimal Value $\hat{y}$ (or Fail)
Binary Reward
$\hat{y} \approx g(x) \rightarrow$ Reward 1
Error/Wrong Answer → Reward 0